# QUASI-FINITENESS OVER A GENERIC POINT VIA GENERIC POINTS OF THE FIBRE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

*Editorial note.* Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

*Editorial note.* The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

*Editorial note.* Read in full under the Lead's 12:4x rule. The proof contains 'It is immediate that (2) implies (3)' - a genuinely immediate implication between two of the listed conditions; the rest of the argument is given in full, so this is not a hint-only proof.

*Allowed standard prerequisites (editorial).*

- Characterisation of quasi-finiteness at a point through the absence of specialisations in the fibre. Lemma \ref{lemma-quasi-finite-at-point-characterize}, cited at proof lines 13318-13319 and 13322-13324. Supplies the criterion used in both directions.

*Source reference.* Original lemma 0BAH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14210–14210. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 01TH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2374–2374. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $X$, $Y$ be schemes. Let $f : X \to Y$ be locally of finite type. Let $X^0$, resp. $Y^0$ denote the set of generic points of irreducible components of $X$, resp. $Y$. Let $\eta \in Y^0$. The following are equivalent*

(1) $f^{-1}(\{\eta\}) \subset X^0$,
(2) *$f$ is quasi-finite at all points lying over $\eta$,*

*Date*: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(3) *$f$ is quasi-finite at all $\xi \in X^0$ lying over $\eta$.*

*Proof.* Condition (1) implies there are no specializations among the points of the fibre $X_\eta$. Hence (2) holds by Lemma 01TH. The implication (2) $\Rightarrow$ (3) is immediate. Since $\eta$ is a generic point of $Y$, the generic points of $X_\eta$ are generic points of $X$. Hence (3) and Lemma 01TH imply the generic points of $X_\eta$ are also closed. Thus all points of $X_\eta$ are generic and we see that (1) holds. □